

NANDINI HAZRA

ASSISTANT PROFESSOR (ADJUNCT) ·

NARODOWE CENTRUM BADAŃ JĄDROWYCH

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CURRENT AFFILIATION

Narodowe Centrum Badań Jądrowych (NCBJ)

PI: PROF. DR. HAB. AGNIESZKA POLLO

- Low surface brightness galaxies in the LSST

Warsaw, Poland

Feb. 2025 - Present

EDUCATION

Gran Sasso Science Institute (GSSI) and Istituto Nazionale di Astrofisica, Osservatorio Astronomico d'Abruzzo (INAF-OAAb)

L'Aquila, Italy

PHD, ASTROPARTICLE PHYSICS

2019 - 2024

- Advisors: Prof. Marica Branchesi, Dr. Michele Cantiello

Indian Institute of Science Education and Research (IISER), Pune

Pune, India

BS-MS (BACHELOR AND MASTER OF SCIENCE), PHYSICS

2013 - 2019

- Thesis Advisor: Prof. Dipankar Bhattacharya (IUCAA Pune)

RESEARCH INTERESTS

- Vera Rubin Observatory, Hyper Suprime-Cam and science with large optical surveys
- Low Surface Brightness (LSB) galaxies
- Distance indicators, in particular Surface Brightness Fluctuations (SBF) and Globular Cluster Luminosity Function (GCLF) with data from HSC-Subaru, HST/ACS, JWST and CFHT.
- Optical transients and their detection strategies with next generation of optical telescopes
- Multi-messenger observations of kilonovae for third generation of gravitational wave observatories, in particular the Einstein Telescope
- Member of the Einstein Telescope collaboration and the Vera Rubin Observatory's science collaborations pertaining to Galaxies, and Transients and Variable Stars

CONFERENCES AND WORKSHOPS

TALKS

(Contributed) Vera Rubin Observatory and Einstein Telescope: kilonova observation strategies to optimize target-of-opportunity follow ups. **XV Einstein Telescope Symposium, Online, May 2025**

(Contributed) Rubin and Einstein Telescope: lessons learned from a kilonova ToO study. **LSST@Europe6, Online, September 2024**

(Contributed) Vera Rubin Observatory and Einstein Telescope: kilonova studies to understand ET detector design. **4th Gravitational Wave Workshop, L'Aquila (Italy), October 2023**

(Invited) Surface Brightness Fluctuations for Rubin: first results from Hyper Suprime Cam (HSC) data. **LSST@Europe5, Poreč (Croatia), September 2023**

(Contributed) Rubin and Einstein Telescope: kilonova studies to understand ET detector design. **LSST@Europe5, Poreč (Croatia), September 2023**

(Contributed) Globular Cluster Luminosity Functions and Surface Brightness Fluctuations in the era of LSST, JWST and Roman. **Roman Science Inspired by Emerging JWST Results, Baltimore (USA), June 2023**

(Contributed) A Globular Cluster Luminosity Function (GCLF) Distance to Hydra I using HST data. **VEGAS collaboration Meeting, Pimonte (Italy), July 2022**

(Informal) Globular Cluster Luminosity Functions with LSST and JWST. **MIAPbP: The Extragalactic Distance Scale and Cosmic Expansion in the Era of Large Surveys and the James Webb Space Telescope, Garching (Germany), July 2023**

POSTERS

(Flash talk + Poster) Surface Brightness Fluctuations in the era of Rubin. **LSST@Europe4, Rome (Italy), September 2023.**
(Poster) Surface Brightness Fluctuations as a distance indicator for Rubin. **LSST Project and Community Workshop (PCW), Tucson (USA), August 2022.**

ATTENDED

2nd Einstein Telescope annual meeting. **Online, November 2023**

LSST's TVS Diversity, Equity and Inclusion workshop on Collective Principles & Community Building. **Online, February 2023**

1st Einstein Telescope annual meeting. **Cascina (Italy), November 2022**

Vera Rubin Observatory's Survey Cadence Optimising Committee (SCOC) workshop. **Online, November 2022**

The Paris-Saclay AstroParticle Symposium. **Paris (France), November 2021**

Scientific Communication in Astronomy School, **Bertinoro (Italy), October 2021**

Kick Off Workshop of the Einstein Telescope Observational Science Board. **Online, September 2021**

Rubin Observatory's Data Preview 0 Workshop. **Online, July 2021**

Ninth International Fermi Symposium. **Online, April 2021**

LSST Broker Workshop. **Online, April 2021**

SIGRAV International School: Gravity of Compact Astrophysical Objects and Gravitational Waves. **Online, February 2021**

11th Einstein Telescope Symposium. **Online, December 2020**

European Astronomical Society Annual Meeting. **Online, July 2020**

FELLOWSHIPS AND GRANTS

July 2023	PhD student grant for the "Extragalactic Distance Scale" workshop , Munich Institute for Astro-, Particle and BioPhysics (MIAPbP)
April-June 2023	Departmental grant for visiting Dr. John P. Blakeslee at the NOIRLab in Tucson (Arizona, USA) to collaborate on the project of Surface Brightness Fluctuations as a distance indicator for Rubin , GSSI L'Aquila and INAF-OAAb
August 2022	LSSTC Enabling Science Grant , LSST Corporation
May-July 2016	DAAD-WISE Summer Fellowship , Deutscher Akademischer Austauschdienst (German Academic Exchange)
2013-2018	DST-Inspire Undergraduate Fellowship , Department of Science and Technology (DST), Government of India

SKILLS

- **Programming Languages:** *Python (advanced), C++ (moderate), C (moderate), Fortran (basic), IDL (basic), Java (basic)*
- **Softwares:** *HEASoft (basic, focused on Xspec), GWFish (basic), experience with both Windows and Linux systems*
- **Languages:** *English (fluent), Bengali (fluent), Hindi (fluent), Marathi (fluent), Italian (advanced)*

ORGANISATION, OUTREACH AND TEACHING

Upcoming	LSST@Europe 7 , SOC	Poznań, Poland
June 2024	Rubin Data Academy workshop , SOC and Tutorial presenter	Online
January 2024	Drafting of Publication Policy for the Galaxies Science Collaboration in Rubin , Working Group Member	Online
June 2023	Rubin Data Preview 0 Virtual Summer School , SOC and Tutorial presenter	Online
August 2022	LSST Project and Community Workshop (PCW) , SOC	Tucson, USA
September 2022	Notte Europea dei Ricercatori (European Night of Researchers) , Volunteer	Teramo, Italy
July 2018	Eklavya Foundation's workshop for training middle and high schools science teachers from public schools in under-served areas across Maharashtra , Teaching Volunteer	Amravati, India
2013-2014	After-school tutoring in science and mathematics for middle-school students in underprivileged localities of Pune , Volunteer	Disha at IISER Pune, India

PUBLICATIONS

- H. Thuruthipilly, K. Lisiecki, Junais, ..., **N. Hazra**, P. Matera et al. 2026
From DES to KiDS: Domain adaptation for cross-survey detections of low-surface-brightness galaxies. *A&A*, **711** (2026) A187
- S. Bisero, S. D. Vergani, E. Loffredo, M. Branchesi, **N. Hazra**, U. Dupletsa, R. I. Anderson 2026.
Multi-messenger observations of binary neutron star mergers: synergies between the next generation gravitational wave interferometers and wide-field, high multiplex spectroscopic facilities. *A&A*, **705**(2026) A54
- G. Riccio, M. Cantiello, R. Habas, **N. Hazra**, G. D'Ago, G. Raimondo, J. P. Blakeslee, J. B. Jensen, M. Mirabile, E. Brocato, M. Brescia, C. M. Raiteri 2025.
FAST-SBF: an automatic procedure for the measurement of Surface Brightness Fluctuations for large sky surveys. *A&A*, **704** (2025) A99
- E. Loffredo, **N. Hazra**, U. Dupletsa, M. Branchesi et al. 2025.
Prospects for optical detections from binary neutron star mergers with the next-generation multi-messenger observatories. *A&A*, **697** (2025) A36
- The Einstein Telescope Collaboration (including **N. Hazra**) 2025.
The Science of the Einstein Telescope. *arXiv:2503.12263*
- V. Mainieri, R.I. Anderson, J. Brinchmann, ..., **N. Hazra**, et al. 2024.
The Wide-field Spectroscopic Telescope (WST) Science White Paper. *arXiv:2403.05398*
- M. Cantiello, J.P. Blakeslee, L. Ferrarese, P. Côté, G. Raimondo, J-C. Cuillandre, P.R. Durrell, S. Gwyn, **N. Hazra**, E.W. Peng, J.C. Roediger, R. Sanchez-Janssen, M. Kurzner 2024.
The Next Generation Virgo Cluster Survey (NGVS. III. A Catalog of Surface Brightness Fluctuation (SBF) Distances and the Three-Dimensional Distribution of Galaxies in the Virgo Cluster. *ApJ*, **Volume 966**, May 2024
- M. Branchesi, M. Maggiore, ..., J. Harms, **N. Hazra**, S. Hild, et al. 2023
Science with the Einstein Telescope: a comparison of different designs. *JCAP*, **Volume 2023**, July 2023
- M. Cadoni, A. P. Sanna, M. Pitzalis, B. Banerjee, R. Murgia, **N. Hazra**, M. Branchesi 2023.
Cosmological coupling of nonsingular black holes. *JCAP*, **Volume 2023**, November 2023
- **N. Hazra**, M. Cantiello, G. Raimondo, M. Mirabile, J.P. Blakeslee, M. Branchesi, E. Brocato 2022.
A study of globular clusters in a lenticular galaxy in Hydra I from deep HST/ACS photometry. *A&A*, **666** (2022), A99.
- Y. Bhargava, **N. Hazra**, A.R. Rao, R. Misra, D. Bhattacharya, J. Roy, Md S. Alam 2022.
Probing the shot behaviour in Cygnus X-1 using simultaneous AstroSat-NICER observation. *MNRAS*, **Volume 512**, Issue 4, June 2022.